Lesson 12: Markov Chains

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14 Markov Chains

14.1 Definition and Transition Matrices

Definition 14.1 (Markov Chain)**.** A sequence of random variables $\{X_t\}_{t\geq 0}$ taking values in a finite or countable state space $\mathcal{S}$ is a *(discrete-time) Markov chain* if for all $t \geq 0$ and all states $s_0, \ldots, s_t, s_{t+1} \in \mathcal{S}$:

$$\mathbb{P}(X_{t+1} = s_{t+1} \mid X_t = s_t, X_{t-1} = s_{t-1}, \ldots, X_0 = s_0) = \mathbb{P}(X_{t+1} = s_{t+1} \mid X_t = s_t). \tag{1}$$

Definition 14.2 (Time-Homogeneous Transition Matrix)**.** A Markov chain is *time-homogeneous* if $\mathbb{P}(X_{t+1} = s' \mid X_t = s) = P(s' \mid s)$ does not depend on t. The *transition matrix* $\mathbf{P} \in \mathbb{R}^{|\mathcal{S}|\times|\mathcal{S}|}$ has entries:

$$P_{ss'} = P(s' \mid s), \quad \text{with } P_{ss'} \geq 0 \text{ and } \sum_{s'\in\mathcal{S}} P_{ss'} = 1 \text{ for all } s. \tag{2}$$

Such a matrix (non-negative entries, rows summing to 1) is called a *stochastic matrix.*

Theorem 14.1 (Chapman–Kolmogorov Equation)**.** *For a time-homogeneous Markov chain with transition matrix* $\mathbf{P}$*:*

$$P^{(n+m)}(s, s') = \sum_{u\in\mathcal{S}} P^{(n)}(s, u)\, P^{(m)}(u, s'), \tag{3}$$

where $P^{(k)}(s, s') = \mathbb{P}(X_{t+k} = s' \mid X_t = s)$ *is the* k*-step transition probability. In matrix notation:* $\mathbf{P}^{n+m} = \mathbf{P}^n\mathbf{P}^m$.

Proof. By the law of total probability and the Markov property:

$$\begin{aligned}
P^{(n+m)}(s, s') &= \mathbb{P}(X_{n+m} = s' \mid X_0 = s) \\
&= \sum_{u\in\mathcal{S}} \mathbb{P}(X_{n+m} = s' \mid X_n = u, X_0 = s)\, \mathbb{P}(X_n = u \mid X_0 = s) \\
&= \sum_{u\in\mathcal{S}} \mathbb{P}(X_{n+m} = s' \mid X_n = u)\, \mathbb{P}(X_n = u \mid X_0 = s) \qquad \text{(Markov property)} \\
&= \sum_{u} P^{(m)}(u, s')\, P^{(n)}(s, u).
\end{aligned}$$

This is precisely the (s, s') entry of the matrix product $\mathbf{P}^n\mathbf{P}^m$. □

14.2 Classification of States

Definition 14.3 (Accessibility and Communication)**.** State s' is *accessible* from s (written $s \to s'$) if $P^{(n)}(s, s') > 0$ for some $n \geq 0$. States s and s' *communicate* (written $s \leftrightarrow s'$) if $s \to s'$ and $s' \to s$.

Proposition 14.2 (Communication is an Equivalence Relation)**.** *The relation* $\leftrightarrow$ *is an equivalence relation on* $\mathcal{S}$*, partitioning the state space into* communicating classes.

Proof. Reflexive: $P^{(0)}(s, s) = 1 > 0$, so $s \to s$.

Symmetric: By definition of $\leftrightarrow$.

Transitive: If $s \leftrightarrow u$ and $u \leftrightarrow s'$, then $s \to u$ in n steps and $u \to s'$ in m steps. By Chapman–Kolmogorov, $P^{(n+m)}(s, s') \geq P^{(n)}(s, u)P^{(m)}(u, s') > 0$, so $s \to s'$. Similarly $s' \to s$. □

Definition 14.4 (Irreducible Chain)**.** A Markov chain is *irreducible* if there is a single communicating class, i.e., $s \to s'$ for all $s, s' \in \mathcal{S}$.

Definition 14.5 (Recurrence and Transience)**.** Let $\tau_s = \inf\{t \geq 1 : X_t = s\}$ be the first return time to state s. State s is:

- *Recurrent* if $\mathbb{P}(\tau_s < \infty \mid X_0 = s) = 1$ (the chain returns to s with probability 1).
- *Transient* if $\mathbb{P}(\tau_s < \infty \mid X_0 = s) < 1$.

Theorem 14.3 (Recurrence Criterion)**.** *State s is recurrent if and only if $\sum_{n=0}^{\infty} P^{(n)}(s,s) = \infty$.*

Proof. Let $N_s = \sum_{t=1}^{\infty} \mathbf{1}_{\{X_t = s\}}$ be the number of returns to s (starting from $X_0 = s$). Then:

$$\mathbb{E}[N_s \mid X_0 = s] = \sum_{t=1}^{\infty} P^{(t)}(s,s). \tag{4}$$

If s is transient, let $p = \mathbb{P}(\tau_s < \infty \mid X_0 = s) < 1$. By the strong Markov property, $\mathbb{P}(N_s \geq k \mid X_0 = s) = p^k$, so N_s is geometric: $\mathbb{E}[N_s \mid X_0 = s] = p/(1-p) < \infty$. Thus $\sum P^{(n)}(s,s) < \infty$.

If s is recurrent, then $p = 1$, so $\mathbb{P}(N_s = \infty \mid X_0 = s) = 1$, giving $\mathbb{E}[N_s \mid X_0 = s] = \infty$. □

Proposition 14.4 (Recurrence is a Class Property)**.** *If s is recurrent and $s \leftrightarrow s'$, then s' is recurrent.*

Proof. Since $s \leftrightarrow s'$, there exist m, n such that $P^{(m)}(s',s) > 0$ and $P^{(n)}(s,s') > 0$. For any k:

$$P^{(m+k+n)}(s',s') \geq P^{(m)}(s',s)\, P^{(k)}(s,s)\, P^{(n)}(s,s'). \tag{5}$$

Let $c = P^{(m)}(s',s) \cdot P^{(n)}(s,s') > 0$. Summing over k:

$$\sum_k P^{(m+k+n)}(s',s') \geq c \sum_k P^{(k)}(s,s) = \infty. \tag{6}$$

So $\sum_t P^{(t)}(s',s') = \infty$, and s' is recurrent by Theorem 14.3. □

14.3 Stationary Distributions

Definition 14.6 (Stationary Distribution)**.** A probability distribution $\pi = (\pi(s))_{s \in \mathcal{S}}$ is *stationary* (or *invariant*) for a Markov chain with transition matrix $\mathbf{P}$ if:

$$\pi(s') = \sum_{s \in \mathcal{S}} \pi(s)\, P(s' \mid s) \quad \text{for all } s' \in \mathcal{S}. \tag{7}$$

In vector notation: $\boldsymbol{\pi}^\top = \boldsymbol{\pi}^\top \mathbf{P}$, i.e., $\boldsymbol{\pi}$ is a left eigenvector of $\mathbf{P}$ with eigenvalue 1.

Definition 14.7 (Aperiodic Chain)**.** The *period* of state s is $d(s) = \gcd\{n \geq 1 : P^{(n)}(s,s) > 0\}$. State s is *aperiodic* if $d(s) = 1$. An irreducible chain is aperiodic if any (equivalently, all) of its states are aperiodic.

Definition 14.8 (Positive Recurrence)**.** A recurrent state s is *positive recurrent* if $\mathbb{E}[\tau_s \mid X_0 = s] < \infty$. An irreducible Markov chain is positive recurrent if all its states are positive recurrent.

Existence and Uniqueness of Stationary Distribution

Theorem 14.5 (Stationary Distribution Theorem)**.** *For a finite, irreducible Markov chain:*

1. *The chain is positive recurrent.*
2. *There exists a unique stationary distribution $\boldsymbol{\pi}$ with $\pi(s) > 0$ for all s.*
3. $\pi(s) = 1/\mathbb{E}[\tau_s \mid X_0 = s]$.

Proof. **Part (1):** In a finite irreducible chain, all states communicate, and at least one state must be recurrent (since the chain cannot escape to infinity). By Proposition 14.4, all states are recurrent. In a finite state space, recurrence implies positive recurrence (the expected return time is finite because the chain has only finitely many states to visit).

Part (2), existence: Define $\pi(s) = 1/\mathbb{E}[\tau_s \mid X_0 = s]$ (which is positive and finite by Part (1)). We verify stationarity. Consider the chain started from $X_0 = s$. The expected number of visits to state s' before returning to s is:

$$\mu(s') = \mathbb{E}\left[\sum_{t=0}^{\tau_s - 1} \mathbf{1}_{\{X_t = s'\}} \;\middle|\; X_0 = s\right]. \tag{8}$$

One can verify that $\mu(s')/\mu(s)$ satisfies the stationarity equation $\boldsymbol{\pi}^\top = \boldsymbol{\pi}^\top \mathbf{P}$ and $\mu(s) = \mathbb{E}[\tau_s \mid X_0 = s]$, so $\pi(s') = \mu(s')/\sum_u \mu(u) = 1/\mathbb{E}[\tau_{s'} \mid X_0 = s']$ after normalization.

Part (2), uniqueness: Suppose $\boldsymbol{\pi}'$ is another stationary distribution. Then $\boldsymbol{\pi}'^\top = \boldsymbol{\pi}'^\top \mathbf{P}^n$ for all n. Since the chain is irreducible and positive recurrent, $P^{(n)}(s, s') \to \pi(s')$ as $n \to \infty$ (if additionally aperiodic; in the periodic case, Cesàro averages converge). Therefore $\pi'(s') = \sum_s \pi'(s)\pi(s') = \pi(s')$. □

Connection to RL

In the TD function approximation setting (Section 6 of the TD article), the stationary distribution d^π of the Markov chain induced by policy π defines the weighting in the Mean Squared Value Error $\overline{\mathrm{VE}}(\mathbf{w}) = \sum_s d^\pi(s)(V^\pi(s) - \hat{V}(s; \mathbf{w}))^2$. The convergence proof of linear TD (Theorem 6.2) relies on $d^\pi(s) > 0$ for all s, which is guaranteed by irreducibility. The ergodic theorem (below) justifies replacing expectations under d^π with time averages along a single trajectory.

14.4 Ergodic Theorem

Definition 14.9 (Ergodic Chain)**.** A finite Markov chain is *ergodic* if it is irreducible and aperiodic.

Convergence Theorem for Ergodic Chains

Theorem 14.6 (Ergodic Theorem for Finite Markov Chains)**.** *If $\{X_t\}$ is an ergodic finite Markov chain with stationary distribution $\boldsymbol{\pi}$, then:*

1. ***Distribution convergence:*** *For any initial distribution, $\mathbb{P}(X_t = s) \to \pi(s)$ as $t \to \infty$.*
2. ***Time-average convergence:*** *For any bounded function $f : \mathcal{S} \to \mathbb{R}$:*

$$\frac{1}{T}\sum_{t=0}^{T-1} f(X_t) \xrightarrow{a.s.} \sum_{s \in \mathcal{S}} \pi(s)\, f(s) = \mathbb{E}_\pi[f]. \tag{9}$$

Proof of Part (1). For an ergodic chain on a finite state space $\mathcal{S} = \{1, \ldots, N\}$, aperiodicity and irreducibility imply that there exists n_0 such that $P^{(n)}(s, s') > 0$ for all s, s' and all $n \geq n_0$.

Let $\boldsymbol{\mu}_t^\top = \boldsymbol{\mu}_0^\top \mathbf{P}^t$ be the distribution at time t. We show $\|\boldsymbol{\mu}_t - \boldsymbol{\pi}\|_1 \to 0$.

Define the *coupling coefficient* $\alpha(\mathbf{P}) = \max_{s,s'} \sum_u |P(u \mid s) - P(u \mid s')|/2$. For a matrix with all positive entries, $\alpha(\mathbf{P}) < 1$. Since $\mathbf{P}^{n_0}$ has all positive entries, $\alpha(\mathbf{P}^{n_0}) = \rho < 1$. One can verify that:

$$\|\boldsymbol{\mu}_{n_0}^\top - \boldsymbol{\pi}^\top\|_1 \leq \rho\, \|\boldsymbol{\mu}_0^\top - \boldsymbol{\pi}^\top\|_1. \tag{10}$$

Iterating: $\|\boldsymbol{\mu}_{kn_0}^\top - \boldsymbol{\pi}^\top\|_1 \leq \rho^k \cdot 2 \to 0$, giving convergence. □

Proof sketch of Part (2). By the strong law of large numbers for Markov chains (which extends the i.i.d. SLLN): for a positive recurrent chain, the fraction of time spent in state s converges a.s. to $\pi(s)$. Since $\frac{1}{T}\sum_t f(X_t) = \sum_s f(s) \cdot \frac{1}{T}\sum_t \mathbf{1}_{\{X_t=s\}}$, the result follows from a.s. convergence of each empirical frequency to $\pi(s)$. □

14.5 Detailed Balance

Definition 14.10 (Detailed Balance / Reversibility)**.** A Markov chain with transition matrix $\mathbf{P}$ satisfies *detailed balance* with respect to a distribution $\boldsymbol{\pi}$ if:

$$\pi(s)\, P(s' \mid s) = \pi(s')\, P(s \mid s') \quad \text{for all } s, s' \in \mathcal{S}. \tag{11}$$

A chain satisfying detailed balance is called *reversible*.

Proposition 14.7 (Detailed Balance Implies Stationarity)**.** *If $\boldsymbol{\pi}$ satisfies detailed balance* (11)*, then $\boldsymbol{\pi}$ is a stationary distribution.*

Proof. Sum both sides of (11) over s:

$$\sum_s \pi(s)\, P(s' \mid s) = \sum_s \pi(s')\, P(s \mid s') = \pi(s') \sum_s P(s \mid s') = \pi(s'). \tag{12}$$

This is the stationarity equation (7). □

Connection to RL

In the linear TD convergence proof (Theorem 6.2 of the TD article), the key matrix $\mathbf{A} = \mathbb{E}_{d^\pi}[\mathbf{x}(S)(\mathbf{x}(S) - \gamma\mathbf{x}(S'))^\top]$ involves the stationary distribution. Reversibility (detailed balance) is *not* assumed — the chain induced by the MDP under a fixed policy is generally non-reversible. The proof instead uses the general ergodic theorem to replace expectations over d^π with time averages.